WHERE VIOLENCE PERSISTS: LATENT SPATIAL FIELDS FOR THE RESIDUAL RISK OF ARMED CONFLICT AND SOCIAL UNREST

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[TODO: Abstract to be written after the application.]

1. Introduction. [TODO: Introduction (not drafted yet).]

2. Data.

2.1. *Conflict and unrest events.* Event data come from the Armed Conflict Location & Event Data Project (ACLED; Raleigh et al., 2010). ACLED records the date, location, actors and type of individual events of political violence and demonstrations, compiled from news media, local partners and other sources. We use events from 1 January 2018 to 31 December 2024 and group ACLED event types into two outcomes: *violent conflict* (battles, explosions/remote violence and violence against civilians) and *social unrest* (riots and protests). Events are aggregated to a regular $0.5^\circ \times 0.5^\circ$ longitude–latitude grid, the resolution of PRIO-GRID (Tollefsen, Strand and Buhaug, 2012), and to calendar weeks. For cell s and week t, let $N_{st}^{(j)}$ be the number of events of type j, and let

$$Y_{st} = \mathbf{1}\Big\{\textstyle\sum_{j\in\mathcal{J}} N_{st}^{(j)} > 0\Big\},$$

where $\mathcal{J}$ is the set of event types of the outcome being modelled. The two outcomes are analysed separately.

The main analysis covers Africa, the Middle East and adjacent regions, the box from 20°W to 65°E and from 36°S to 45°N, with 13,719 cells in 93 countries. We call it the AME region. Besides Africa and the Middle East, the box includes Mediterranean Europe, the Caucasus, parts of Central Asia and Afghanistan. ACLED coverage of these areas has a different history. Among the countries in the box, Italy, Spain, Portugal, France, Malta and Andorra are covered by ACLED only from 2020 (as are several small islands of the Indian and Atlantic oceans), so they contribute three of the five training years, 2020–2022, which include the period of COVID-19 restrictions.

Event weeks are rare. In 2023–2024, over the covered cell-weeks of the AME region with complete history, 2.4% of cell-weeks contain at least one conflict event and 2.7% at least one unrest event. Over all covered cell-weeks worldwide with complete climate records the shares are 1.6% and 2.7%.

Figure 1 shows where and when events are recorded in the AME region. Over 2018–2024, 2.3% of covered cell-weeks in this region contain at least one conflict event and 2.3% at least one unrest event. Worldwide, the shares are 1.3% and 2.1%. Events are concentrated in space:

- a third of the cells in the region (33.6% for conflict, 35.7% for unrest) record any event at all;

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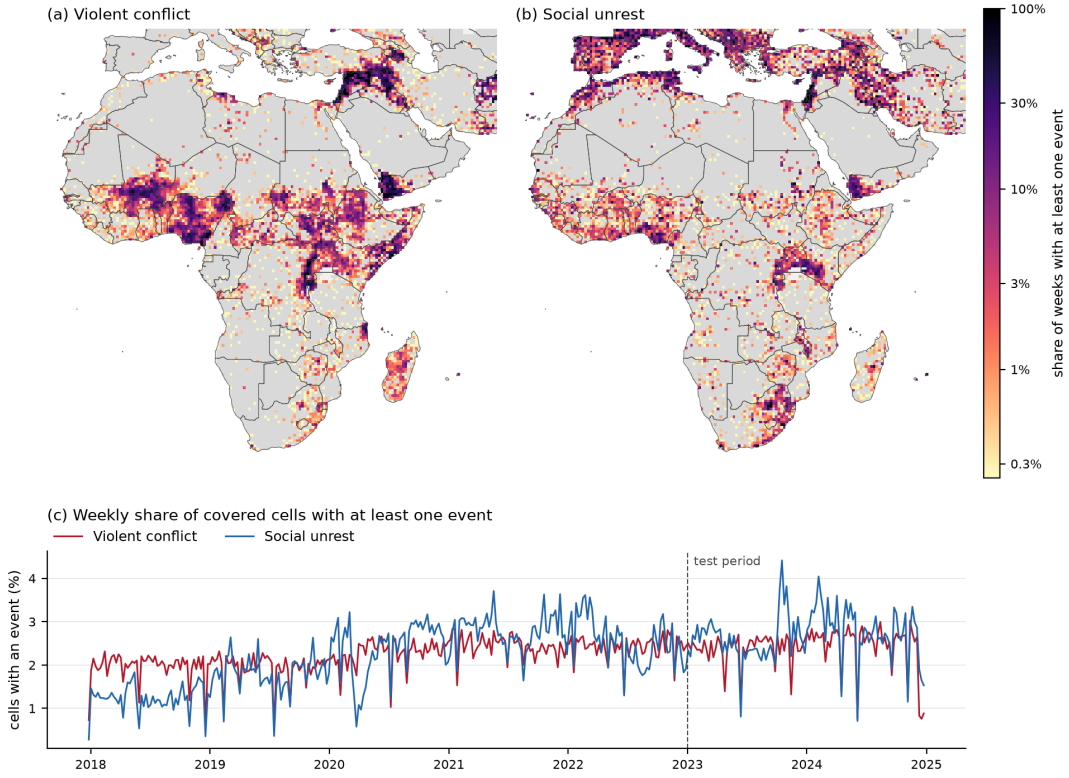


FIG 1. *ACLED events on the* 0.5° *grid in the AME region (Africa, the Middle East and adjacent regions), 2018–2024, over covered cell-weeks. (a, b) Share of weeks with at least one event in each cell, for violent conflict and social unrest (logarithmic colour scale); grey cells have no event week. (c) Weekly share of covered cells in the region with at least one event. The dashed line marks the start of the test period.*

- the 1% of cells with the most event weeks account for 30% of all conflict event weeks and 27% of all unrest event weeks;
- fewer than 1% of cells record events in more than half of all weeks. These cells, for example in Syria, Yemen, northeastern Nigeria and eastern Democratic Republic of the Congo, are one reason why recent history is such a strong predictor (Section 2.2).

The two outcomes have different geographies. Conflict forms large contiguous regions: the central Sahel and Lake Chad basin, the Sudans and Ethiopia, Somalia, eastern Congo, Yemen and Syria–Iraq. Unrest is densest in Mediterranean Europe, North Africa, Turkey, Iran, South Africa and the Great Lakes region, where population and media coverage are high. The weekly share of cells with an event rises over the period. For conflict it goes from about 2% in 2018 to about 2.5% in 2024, and for unrest from about 1.3% to between 2% and 4%. The test period is therefore not exchangeable with the early training years. The series also has isolated weeks with far fewer events than the neighbouring weeks. We checked whether they arise in our construction of the table. No week is split between the partitions of the source table, every cell-week appears once, and the number of cells is constant in the weeks around each of them. The dips are therefore present in the weekly counts of the source partitions, and we keep them as recorded. [TODO: Sensitivity of the main results to excluding these weeks.]

The grid consists of the 0.5° cells that intersect a country, excluding cells without a terrestrial biome. Cell-weeks in a country before its ACLED coverage start are unobserved, not event free, and are removed. Since the probability that an event is recorded depends on media and partner density, the outcome combines occurrence and reporting (Weidmann,

2016); the latent fields of Section 3 absorb spatially smooth reporting differences together with unmeasured drivers of risk, and we interpret them accordingly. The construction of the table, including the repair of duplicated cell-weeks in the original pipeline, is described in Section S1.1 of the Supplementary Material.

2.2. *Recent event history.* For each cell and week we compute the number of events of each type in the four preceding weeks, $L_{st}^{(j)} = \sum_{l=1}^{4} N_{s,t-l}^{(j)}$, using only past weeks of the same cell. In the models of Section 3 the history enters through $h_{st} = \log\bigl(1 + \sum_{j \in \mathcal{J}} L_{st}^{(j)}\bigr)$ for the outcome being modelled, and through the corresponding quantity for the other outcome as an additional covariate. Cell-weeks whose four-week history is incomplete, at the start of the series and of each country's coverage, are dropped from the parametric latent models. The boosting models, and the boosting offset of the hybrid models, keep these rows and receive the incomplete history as missing.

2.3. *Environmental, socioeconomic and national covariates.* Covariates are listed in Table S1. They vary at three resolutions, which is central to the identification arguments of Section S2.7:

1. *cell–week*: daily minimum and maximum temperature, heat index and wet-bulb globe temperature (CHIRTS), precipitation (CHIRPS), $PM_{2.5}$ (CAMS) and a drought index (Global Drought Observatory);
2. *cell*: population (GHSL), river length and lake area (Natural Earth), malaria risk (Malaria Atlas Project), terrestrial biome (RESOLVE Ecoregions) and, in the boosting models only, an indicator of $|\text{latitude}| < 30°$;
3. *country–year*: GDP per capita, access to electricity, the Environmental Performance Index (EPI), the V-Dem liberal democracy index, and per-capita greenhouse gas emissions, material footprint and ecological footprint.

Food insecurity, the population in each IPC phase, is available for subnational regions and periods and is apportioned to cells by population share.

Climate variables enter as their means over the four weeks before the current week, as standardised anomalies with respect to a cell- and calendar-month-specific climatology of the training years 2018–2022. Covariates are standardised with training-period means; missing values are set to zero and a missingness indicator enters for each national indicator and for IPC. Missingness is substantial (IPC phases 62.7% of rows, $PM_{2.5}$ 55.5%, GDP 3.6%) and unlikely to be at random. Section S1.2 of the Supplementary Material lists the covariates and sources (Table S1) and details the anomalies and missing values.

3. Methods.

3.1. *A latent Gaussian model for event occurrence.* Let $\mathcal{S}$ be the set of n grid cells, $c(s)$ the country and $b(s)$ the biome of cell s, and $t = 1, \dots, T$ the weeks. Conditionally on a set of latent effects, the outcomes are independent Bernoulli variables, $Y_{st} \mid \pi_{st} \sim \text{Bernoulli}(\pi_{st})$, with

$$\text{logit}\, \pi_{st} = \boldsymbol{x}_{st}^{\top}\boldsymbol{\beta} + \delta\, h_{st} + \sum_{k \in \mathcal{K}} z_{k,st}\, \beta_k(s) + \boldsymbol{z}_{c(s),t}^{\top}\boldsymbol{\alpha} + u_{c(s)} + w(s) + e_s + v(s,t). \tag{3.1}$$

Here h_{st} is the recent event history of Section 2.2, so δ measures state dependence, and $\boldsymbol{x}_{st}$ holds the cell and cell-week covariates with fixed effects, including the missingness indicators, biome and year effects. The terms $z_{k,st}\beta_k(s)$ are spatially varying climate effects and

$\boldsymbol{z}_{ct}^{\top}\boldsymbol{\alpha}$ the national indicators; both are used only in the univariate analyses of the Supplementary Material (Sections S2.6 and S2.7). The country effects $u_c \sim \mathcal{N}(0, \sigma_u^2)$ and cell effects $e_s \sim \mathcal{N}(0, \sigma_e^2)$ are independent, w is a Matérn field on the sphere (Section 3.2), and v is a space-time field (Section 3.3). The four latent terms capture persistent *recorded* risk at four scales, each mixing occurrence with reporting: national institutions and reporting regimes (u), regional hotspots that cross borders (w), single cells such as a town or a border crossing (e), and slowly evolving episodes (v).

Model (3.1) is a latent Gaussian model (Rue, Martino and Chopin, 2009). Stacking all latent quantities into $\boldsymbol{\xi} = (\boldsymbol{\beta}, \delta, \boldsymbol{\gamma}, \boldsymbol{\alpha}, \boldsymbol{u}, \boldsymbol{w}, \boldsymbol{e}, \boldsymbol{\eta}, \boldsymbol{v})$, the linear predictor is $\boldsymbol{\eta} = \boldsymbol{o} + \boldsymbol{A}\boldsymbol{\xi}$ for a sparse design matrix $\boldsymbol{A}$ and an optional offset $\boldsymbol{o}$, and $\boldsymbol{\xi} \mid \boldsymbol{\theta} \sim \mathcal{N}\{\boldsymbol{0}, \boldsymbol{Q}(\boldsymbol{\theta})^{-1}\}$ with a sparse, block-diagonal precision whose blocks are listed in Table S7 of the Supplementary Material (Rue and Held, 2005). Fixed effects receive vague $\mathcal{N}(0, 10^4)$ priors and are integrated out together with the random effects.

3.2. *Matérn fields on the sphere via a finite-volume SPDE.* Following Lindgren, Rue and Lindström (2011), a Gaussian field w on $\mathbb{S}^2$ with Matérn-like covariance is defined as the stationary solution of

$$\tau\,(\kappa^2 - \Delta_{\mathbb{S}^2})\,w(s) = \mathcal{W}(s), \qquad s \in \mathbb{S}^2, \tag{3.2}$$

where $\Delta_{\mathbb{S}^2}$ is the Laplace–Beltrami operator and $\mathcal{W}$ is Gaussian white noise (see also Lindgren, Bolin and Rue, 2022). This is the $\alpha = 2$ case, which corresponds to Matérn smoothness $\nu = 1$ in the planar limit. The field then has practical range $\rho = \sqrt{8}/\kappa$ and marginal variance $\sigma^2 = 1/(4\pi\kappa^2\tau^2)$, provided ρ is small relative to the Earth's radius R. We parameterise the field by (ρ, σ), in kilometres and logit units.

The grid cells are natural control volumes, so instead of a triangulated mesh we discretise (3.2) by finite volumes on the longitude–latitude lattice itself (Fuglstad et al., 2015). With $\boldsymbol{C}$ the diagonal matrix of cell areas and $\boldsymbol{G}$ the graph Laplacian of the lattice with edge weights given by the ratio of shared edge length to centre distance, $\boldsymbol{K} = \kappa^2\boldsymbol{C} + \boldsymbol{G}$ and

$$\boldsymbol{w} \sim \mathcal{N}\{\boldsymbol{0}, \boldsymbol{Q}_s^{-1}\}, \qquad \boldsymbol{Q}_s(\rho, \sigma) = \tau^2\boldsymbol{K}\boldsymbol{C}^{-1}\boldsymbol{K}, \tag{3.3}$$

the lumped-mass finite-volume analogue of the finite-element representation of Lindgren, Rue and Lindström (2011), with at most 13 non-zeros per row. It accounts for the shrinking east–west spacing towards the poles and can be restricted to land cells. To keep correlation from leaking across narrow seas, for example between Yemen and Somalia, and to avoid inflated variance at the coast, we use the barrier model of Bakka et al. (2019) on a lattice extended by six rings of cells, with range 0.5ρ in sea cells. Section S2.1 of the Supplementary Material gives the discretisation and the barrier construction, and checks them.

Priors.. We use penalised-complexity priors (Simpson et al., 2017; Fuglstad et al., 2019): $P(\rho < \rho_0) = 0.05$ and $P(\sigma > \sigma_0) = 0.01$, with $\rho_0 = 100$ km (about two grid cells) and $\sigma_0 = 1$. The PC prior of Fuglstad et al. (2019) is derived for Matérn fields on $\mathbb{R}^2$; we use it as an approximation on the sphere, which is adequate for ranges far below R. The country and cell effects have PC priors with $P(\sigma_u > 1) = 0.01$ and $P(\sigma_e > 1) = 0.01$.

3.3. *Space-time persistence.* The space-time term is a separable field that evolves as a first-order autoregression of Matérn innovations,

$$\boldsymbol{v}_\tau = \phi\,\boldsymbol{v}_{\tau-1} + \sqrt{1-\phi^2}\,\boldsymbol{\epsilon}_\tau, \qquad \boldsymbol{\epsilon}_\tau \sim \mathcal{N}\{\boldsymbol{0}, \boldsymbol{Q}_s(\rho_v, \sigma_v)^{-1}\},$$

(Cameletti et al., 2013), so that $\boldsymbol{v}$ has precision $\boldsymbol{Q}_{\mathrm{AR}}(\phi) \otimes \boldsymbol{Q}_s(\rho_v, \sigma_v)$, where $\boldsymbol{Q}_{\mathrm{AR}}$ is the tridiagonal precision of a unit-variance AR(1) process. To control cost, τ indexes blocks of Δ weeks; in the application v uses $\Delta = 13$ weeks and is defined on the 1° cells that contain observed 0.5° cells (Section S2.2). The prior on $\operatorname{atanh}\phi$ is $\mathcal{N}(0, 1.5^2)$.

3.4. *A joint model of conflict and unrest.* Recent unrest is associated with later conflict in the same and in neighbouring places. This association has two readings. Unrest may escalate into armed conflict, for example through repression (Ritter and Conrad, 2016; Rød and Weidmann, 2023; Ryckman, 2020; Chenoweth, Perkoski and Kang, 2017). Alternatively, both outcomes may be driven by the same latent episodes, such as a period of political crisis in a region, or a period of intense media attention to it, without any effect of one on the other. Forecasting studies use unrest histories and their spatial lags as predictors of conflict (Rød, Hegre and Leis, 2025), and diffusion models add spatial lags of past events to the linear predictor (Racek, Thurner and Kauermann, 2026). Neither separates the two readings. We do so by modelling both outcomes jointly, with a space-time field that is common to them.

Let $j \in \{\mathrm{C}, \mathrm{U}\}$ index conflict and unrest, and let Y_{jst} be the indicator of a recorded event of type j in cell s and week t. The history h_{jst} is $\log(1+n)$, where n is the number of events of type j in cell s in weeks $t-4, \ldots, t-1$, and $\bar{h}_{jst}$ is the mean of $h_{js't}$ over the rook neighbours s' of s, standardised on the training rows. Conditionally on the latent effects, the Y_{jst} are independent Bernoulli variables with

$$
\begin{aligned}
\operatorname{logit} \pi_{\mathrm{C}st} &= \boldsymbol{x}_{st}^{\top}\boldsymbol{\beta}_{\mathrm{C}} + \delta_{\mathrm{C}}\, h_{\mathrm{C}st} + \kappa\, h_{\mathrm{U}st} + \delta_{\mathrm{C}}^{\mathrm{nb}}\, \bar{h}_{\mathrm{C}st} + \kappa^{\mathrm{nb}}\, \bar{h}_{\mathrm{U}st} \\
&\quad + u_{\mathrm{C},c(s)} + e_{\mathrm{C}}(s) + w_{\mathrm{C}}(s) + v(s,t), \\
\operatorname{logit} \pi_{\mathrm{U}st} &= \boldsymbol{x}_{st}^{\top}\boldsymbol{\beta}_{\mathrm{U}} + \delta_{\mathrm{U}}\, h_{\mathrm{U}st} + \kappa_{\mathrm{U}}\, h_{\mathrm{C}st} + \delta_{\mathrm{U}}^{\mathrm{nb}}\, \bar{h}_{\mathrm{U}st} + \kappa_{\mathrm{U}}^{\mathrm{nb}}\, \bar{h}_{\mathrm{C}st} \\
&\quad + u_{\mathrm{U},c(s)} + e_{\mathrm{U}}(s) + w_{\mathrm{U}}(s) + \lambda\, v(s,t).
\end{aligned}
\tag{3.4}
$$

The vector $\boldsymbol{x}_{st}$ holds the covariates of model (3.1) that have fixed effects, including the initial-conditions term, the national indicators and the missingness indicators. All fixed effects are specific to the outcome. There are no spatially varying coefficients. For each outcome, $u_{j,c}$ are independent country effects, e_j independent cell effects and w_j a static Matérn field with sea barriers (Remark 1). The two sets of these terms are independent of each other, with their own hyperparameters. The space-time field v is the AR(1) $\otimes$ Matérn field of Section 3.3. It enters the conflict predictor with loading one and the unrest predictor with loading λ. The symbol κ here denotes the escalation coefficient. It is unrelated to the SPDE parameter of Section 3.2.

A loading in the design would make $\boldsymbol{A}$ depend on the hyperparameters; as in the copy construction of R-INLA we instead introduce $\boldsymbol{b} = \lambda\boldsymbol{a} + \boldsymbol{\varepsilon}$, with $\boldsymbol{a}$ the values of v and $\boldsymbol{\varepsilon} \sim \mathcal{N}(\boldsymbol{0}, 10^{-4}\boldsymbol{I})$, let unrest rows load on $\boldsymbol{b}$, and put λ in the prior precision of $(\boldsymbol{a}, \boldsymbol{b})$, whose log-determinant does not depend on λ (Section S2.3). The loading has a $\mathcal{N}(0, 2^2)$ prior; the loading of conflict is fixed at one, which identifies the sign of λ. With $\lambda = 0$ the conflict equation is the univariate model.

The estimand.. The coefficient κ is the association between recent unrest in a cell and conflict in the same cell, conditional on the shared episode $v(s,t)$ and on the other terms of (3.4); κ^{nb} is the same for unrest in the neighbouring cells. We call κ escalation conditional on shared episodes. Like δ (Section S2.5), it is a conditional association of recorded events, not a causal effect. Univariate models of conflict estimate a different quantity, even when they include their own space-time field. Suppose shared episodes drive both outcomes. Unrest in the previous four weeks was then generated in part by the episode at that time, and the episode persists, so $h_{\mathrm{U}st}$ is correlated with $v(s,t)$. A univariate conflict model treats its latent field as independent of the covariates. It estimates the field from conflict outcomes alone, and each latent value is informed by few Bernoulli observations, so the estimate is noisy and shrunk towards zero. The part of the episode that the field does not capture is still visible in $h_{\mathrm{U}st}$, which is an observed, if noisy, proxy of the episode. The coefficient of $h_{\mathrm{U}st}$ then absorbs it, and $\hat{\kappa}$ is biased upward. This is the random-effects exogeneity problem of Sections S2.5

and S2.7 in another form. In the joint model the unrest outcomes enter the likelihood of v. The episode is estimated from both outcomes, and the dependence of $h_{\mathrm{U}st}$ on it is part of the model, because $h_{\mathrm{U}st}$ is built from earlier values of a modelled outcome. Section 4.1 checks this argument by simulation.

Identification and limits.. The shared field is identified from the joint movement of the two outcomes at its own resolution, beyond what the fixed effects and the outcome-specific static terms explain. The coefficient κ is identified from variation in four-week unrest histories within that resolution. Two consequences follow. First, escalation that unfolds over several months and over a region the size of the range of v cannot be told apart from a shared episode, and is attributed to v. The estimate $\hat{\kappa}$ is therefore the short-run, local part of escalation, and it can understate escalation at longer time scales. Second, shared episodes that are shorter than Δ weeks or finer than the lattice of v remain in κ. Third, episodes that affect unrest only are also forced into v, because the model has no unrest-specific space-time field. Since v enters the conflict predictor with loading one, conditioning on v then removes part of the variation in $h_{\mathrm{U}st}$ through which escalation would act. This biases $\hat{\kappa}$ towards zero, even when escalation is real. The drop from the univariate to the joint estimate is therefore consistent both with the removal of confounding by shared episodes and with this attenuation. To separate the two, we fit an extended joint model with an additional unrest-specific space-time field of the same form as v, and we extend the simulation of Section 4.1 with scenarios that contain unrest-only episodes, with and without escalation. In those scenarios the attenuation did not occur; the unrest-only episodes instead inflated the estimated loading (Section 4.1).

The shared component mixes common shocks to occurrence with common shocks to reporting, such as a period of media attention to a place (Weidmann, 2016; Hellmeier, Weidmann and Rød, 2018). With a single source of event data the two cannot be separated (Cook et al., 2017), so we read κ as conditional on shared episodes of either kind. Model (3.4) is a linear model of coregionalisation (Gelfand et al., 2004); unlike multivariate Hawkes models of event data (Jun and Cook, 2024; Browning et al., 2026), which explain co-occurrence by triggering, it adds a shared stochastic background to cross-lagged terms (Section S2.3).

3.5. *Inference.* For fixed hyperparameters $\boldsymbol{\theta}$, the posterior mode $\boldsymbol{\xi}^*$ of the latent vector is found by damped Newton iterations, and the Laplace approximation to the marginal likelihood is

(3.5)
$$\log \tilde{p}(\boldsymbol{y} \mid \boldsymbol{\theta}) = \log p(\boldsymbol{y} \mid \boldsymbol{\xi}^*) - \tfrac{1}{2}\boldsymbol{\xi}^{*\top}\boldsymbol{Q}\boldsymbol{\xi}^* + \tfrac{1}{2}\log|\boldsymbol{Q}| - \tfrac{1}{2}\log|\boldsymbol{Q}+\boldsymbol{H}|, \qquad \boldsymbol{H} = \boldsymbol{A}^\top \boldsymbol{W}^* \boldsymbol{A}.$$

The hyperparameters are set to the maximiser $\hat{\boldsymbol{\theta}}$ of $\log \tilde{p}(\boldsymbol{y} \mid \boldsymbol{\theta}) + \log p(\boldsymbol{\theta})$, found by L-BFGS-B with finite-difference gradients, whose evaluations run in parallel. Given $\hat{\boldsymbol{\theta}}$, the latent posterior is approximated by $\mathcal{N}\{\boldsymbol{\xi}^*, (\boldsymbol{Q}+\boldsymbol{H})^{-1}\}$, sampled by reusing the sparse Cholesky factorisation computed at the mode. This is an empirical Bayes version of INLA (Rue, Martino and Chopin, 2009), similar in spirit to TMB (Kristensen et al., 2016). The intervals ignore the uncertainty in $\hat{\boldsymbol{\theta}}$, and in simulations with a space-time field the plug-in standard errors of lag coefficients are too small, by a factor of up to 1.2 for $\hat{\kappa}$ (Section 4.1); intervals that only just exclude zero should be read with this in mind. Section S2.4 of the Supplementary Material describes the computation.

Run times and approximations.. The full fits of the joint models, on one node with 16 cores (AMD EPYC) and eight parallel gradient workers, take hours. The joint model (3.4) took 4.5 hours for West Africa and the Sahel (906,810 cell-weeks per outcome) and 6.4 hours for Africa and the Middle East (4,921,758 cell-weeks per outcome). The extended model with an unrest-specific space-time field took 8.2 hours for West Africa from the default starting

values. All estimates and maps reported in Sections 5.1 come from full fits. The repeated refits of the hold-out experiments, one per fold, use three approximations, each checked against a full fit.

First, *warm starts*: an extended model starts from the fitted hyperparameters of the model it extends, and its new components from their defaults. For the extended model in West Africa this reached the same optimum as the cold start in 1.4 hours instead of 8.2. Second, *looser outer tolerances* for L-BFGS-B (relative change of the objective 10^{-7}, projected gradient 10^{-3}). In a simulation replicate of S5 these moved the hyperparameters by at most 0.3% and $\hat{\kappa}$ by 10^{-4}, and the fit was 1.25 times faster. Third, *case-control subsampling of non-events*. Events are rare, so most of the 10^6–10^7 rows are zeros, and the cost of every evaluation grows with the number of rows. For the hyperparameters only, we keep every event and a random fraction r of the non-events of each outcome. For the retained rows,

$$\operatorname{logit} P(y_{st} = 1 \mid \text{retained}) = \eta_{st} - \log r,$$

so adding the offset $-\log r$ gives the exact likelihood of the subsample under the same model (Prentice and Pyke, 1979). Only $\hat{\boldsymbol{\theta}}$ is computed on the subsample. The latent posterior, and therefore every coefficient and prediction, is then computed on all rows at $\hat{\boldsymbol{\theta}}$, with one Newton solve. For the joint model in West Africa, $r = 0.25$ made the fit 5.6 times faster (49 minutes instead of 4.5 hours). The Laplace log marginal likelihood of all rows at the subsample estimate was within 0.6 of its maximum, $\hat{\kappa}$ was 0.047 against 0.044 (0.1 standard errors), every other coefficient moved by less than 0.1 standard errors, and the test log score changed by 0.06%. With $r = 0.10$ the fit was 6.0 times faster but the loading moved from 0.87 to 0.77 and $\hat{\kappa}$ by 0.3 standard errors, so we use $r = 0.25$ [TODO: add the same check for Africa and the Middle East (job 1711777.3) and for one hold-out fold refitted in full].

Prediction.. For a new cell-week with design row $\boldsymbol{a}$ and offset o, the posterior predictive probability is $\mathbb{E}\{\operatorname{expit}(o + \boldsymbol{a}^\top \boldsymbol{\xi})\} \approx \operatorname{expit}\{m/\sqrt{1 + \pi s^2/8}\}$, where m and s^2 are the posterior mean and variance of $o + \boldsymbol{a}^\top \boldsymbol{\xi}$; s^2 is estimated from posterior samples. Test years beyond the training period take the year effect of the last training year.

4. Simulation studies. Two simulation studies address what the application cannot settle on its own: whether an association between recent unrest and conflict reflects escalation or shared latent episodes, and whether a joint model of the two outcomes separates them (S5); and whether the joint model predicts one outcome better where it is not observed but the other outcome is (S6). Studies S0–S4 of the Supplementary Material (Section S3) check the finite-volume SPDE and the Laplace engine (S0) and the univariate models (S1–S4). All code is in the supplementary material.

4.1. *S5: escalation versus shared episodes.* In the data, recent unrest in a cell predicts conflict there (Section 5.1). This can mean that unrest escalates into violence. It can also mean that both outcomes respond to the same latent episodes, such as a local political crisis, so that recent unrest is only a proxy for the current state of the episode. S5 asks whether univariate models with latent fields can tell the two apart, and whether the joint model of Section 3.4 can.

Design.. The domain is a 16×14 block of 0.5° cells (224 cells), with 12 burn-in weeks, 104 training weeks and 26 test weeks. Conflict C_{st} and unrest U_{st} are generated jointly from

$$\operatorname{logit} P(C_{st} = 1) = b_c + d_c L^c_{st} + \kappa L^u_{st} + a(s) + w_c(s) + e(s,t),$$
$$\operatorname{logit} P(U_{st} = 1) = b_u + d_u L^u_{st} + 0.8\, a(s) + w_u(s) + \lambda\, e(s,t),$$

where L^c_{st} and L^u_{st} are $\log(1+\text{events in the previous four weeks})$ of each outcome, $d_c = d_u = 0.3$, $b_c = -3.9$ and $b_u = -3.6$. The static field a is shared by both outcomes (SD 0.6, range 400 km). The fields w_c (SD 0.5, range 300 km) and w_u (SD 0.5, range 150 km) are specific to each outcome. The shared episodes e follow an AR(1) process in time with coefficient 0.95 per week, whose innovations are Matérn fields with range 300 km, scaled to a marginal SD of 0.8. They enter unrest with loading $\lambda = 0.8$. As in the other simulations, every field is drawn from the continuous Matérn covariance, not from the lattice GMRF. The escalation effect is $\kappa = 0$ (no escalation) or $\kappa = 0.4$ (escalation). Shared episodes raise both outcomes together, so recent unrest predicts conflict even when $\kappa = 0$. In two further scenarios, unrest also has episodes of its own: a second AR(1) process with the same coefficient, Matérn innovations of range 200 km and marginal SD 0.6, which enters unrest only. Without these unrest-only episodes, the data-generating process has a single shared space-time field, so the joint model with one shared field is correctly specified in this respect. With them, that model must absorb the unrest-only episodes into the shared field, which can bias $\hat{\kappa}$ towards zero. Event weeks make up about 3.4% (no escalation) and 3.7% (escalation) of conflict rows and 3.9% of unrest rows.

We compare five estimators of κ, fitted to the training weeks, each with 20 replicates per scenario:

- logistic regression of conflict on L^c and L^u;
- the same with a static Matérn field;
- the same with a static field and a space-time AR(1) field at four-week steps (Section 3.3);
- the joint model of both outcomes (Section 3.4), with outcome-specific intercepts, lags and static fields, and a space-time field at four-week steps shared by the two outcomes, which enters unrest with an estimated loading λ;
- the extended joint model, which adds a space-time field for unrest alone at the same four-week steps.

The joint model is not the data-generating process. It has no shared static field, so a must be absorbed by the two outcome-specific fields, and its space-time field is piecewise constant over four weeks while e changes weekly. The application differs from this design: it uses steps of 13 weeks on a 1° lattice, cross-lags in both directions and neighbour histories of both outcomes (Section 5.1). Prediction is studied in S6 (Section 4.2).

Results.. Figure 2 summarises the estimates, and Table S17 of the Supplementary Material gives the full results. Monte Carlo SEs of coverage and rejection rates are at most 0.11 with 20 replicates.

- *No escalation.* All three univariate models report escalation that does not exist. Logistic regression gives $\hat{\kappa} = 0.70$ (empirical SD 0.19), the static field 0.43 (SD 0.16) and the space-time field 0.27 (SD 0.16). They reject $\kappa = 0$ in 95%, 85% and 55% of replicates. The joint model gives 0.04 (SD 0.17, Monte Carlo SE of the mean 0.04), with coverage 0.90 (Monte Carlo SE 0.07) and a false rejection rate of 0.10.
- *Escalation.* The univariate models overestimate $\kappa = 0.4$ (1.03, 0.79 and 0.66), with coverage 0.00, 0.10 and 0.50. The joint model gives 0.43 (SD 0.11, Monte Carlo SE of the mean 0.025), with coverage 0.95 (Monte Carlo SE 0.05), and it rejects $\kappa = 0$ in 95% of replicates.
- *Unrest-only episodes.* The univariate models are biased as before ($\hat{\kappa} = 0.68$, 0.41 and 0.28 without escalation; 1.02, 0.79 and 0.68 with escalation). The attenuation feared for the joint model with one shared field does not appear: it gives -0.01 (SD 0.17, coverage 0.85) without escalation and 0.39 (SD 0.12, coverage 0.90) with escalation. Its loading absorbs the unrest-only episodes instead: $\hat{\lambda}$ has mean 1.17–1.19 (SD 0.29–0.32) against a true 0.8,

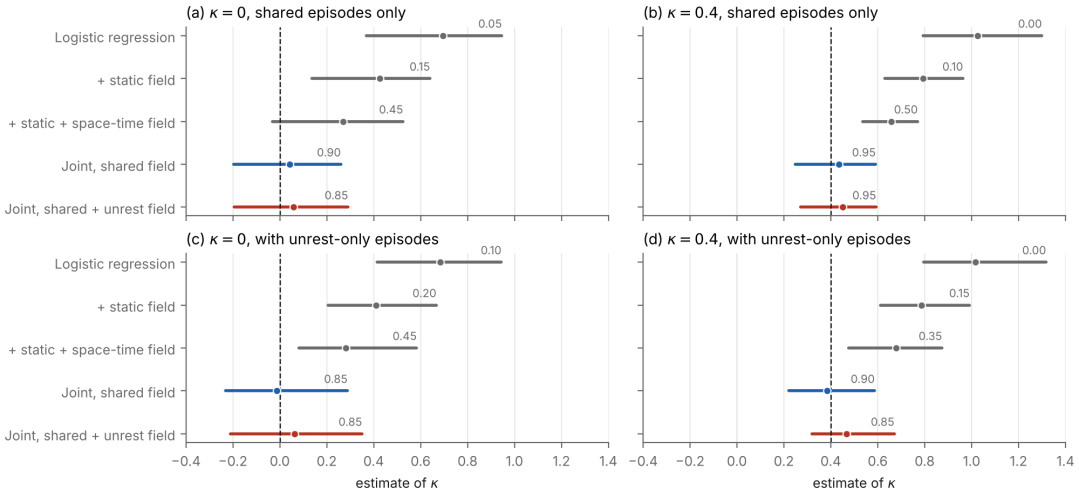


FIG 2. *S5: estimates of the escalation effect κ of recent unrest on conflict, over 20 replicates per scenario, without (a, b) and with (c, d) unrest-only episodes. Dots are means and lines span the 5th to 95th percentiles of the estimates across replicates; the number above each line is the coverage of the nominal 95% intervals for κ. Grey: univariate models of conflict; blue: the joint model with one shared space-time field; red: the extended joint model with an additional space-time field for unrest. The dashed line marks the true value. Full results are in Table S17 of the Supplementary Material.*

and $\hat{d}_c$ rises to 0.44–0.47. The extended model recovers the loading (0.81) and d_c (0.35–0.37), and gives $\hat{\kappa} = 0.06$ and 0.47, with coverage 0.85 in both scenarios.

In the scenarios with shared episodes only, the extended model gives 0.06 and 0.45 (coverage 0.85 and 0.95), close to the model with one shared field. A space-time field for conflict alone removes most of the bias in the own-lag coefficient d_c (0.37–0.38 against 0.3, compared with 1.52–1.54 for logistic regression and 0.80–0.82 with the static field), but not the bias in κ. A plausible explanation is that conflict events are rare, so a field estimated from conflict alone tracks the episodes poorly, and recent unrest still carries information about the current episode. The joint model estimates the field from both outcomes. Its estimate of d_c is 0.36–0.38.

The joint model also recovers the shared field. The estimated loading has mean 0.83 in both scenarios (SD 0.14 without and 0.15 with escalation) against a true 0.8, and the range of the shared field has mean 315 km (SD 40–44) against 300 km. As in S3 (Section S3.5), the standard error of the lag coefficient is too small when a space-time field is fitted under no escalation: the empirical SD of $\hat{\kappa}$ is 1.20 times the mean SE (0.17 against 0.14). Under escalation the ratio is 0.89. With 20 replicates these ratios are imprecise, but together with S3 they indicate that plug-in intervals for κ near zero are too short.

These results hold for a single shared episode process whose loading is constant in space, with one direction of escalation. Faster episodes, a loading that varies by region, escalation that runs in both directions, and the 13-week steps of the application are not covered. Unrest-only episodes were tested at one strength (SD 0.6, range 200 km); with them, the model with one shared field still recovers κ but misestimates λ and d_c, so its loading should not be interpreted when unrest may have episodes of its own.

4.2. *S6: prediction where one outcome is unobserved.* When both outcomes are observed everywhere, the joint model cannot be expected to predict either one better than univariate models that use both histories as covariates: the shared field is then informed by each outcome's own data in the same way, and in S5 the joint and univariate forecasts of conflict differ

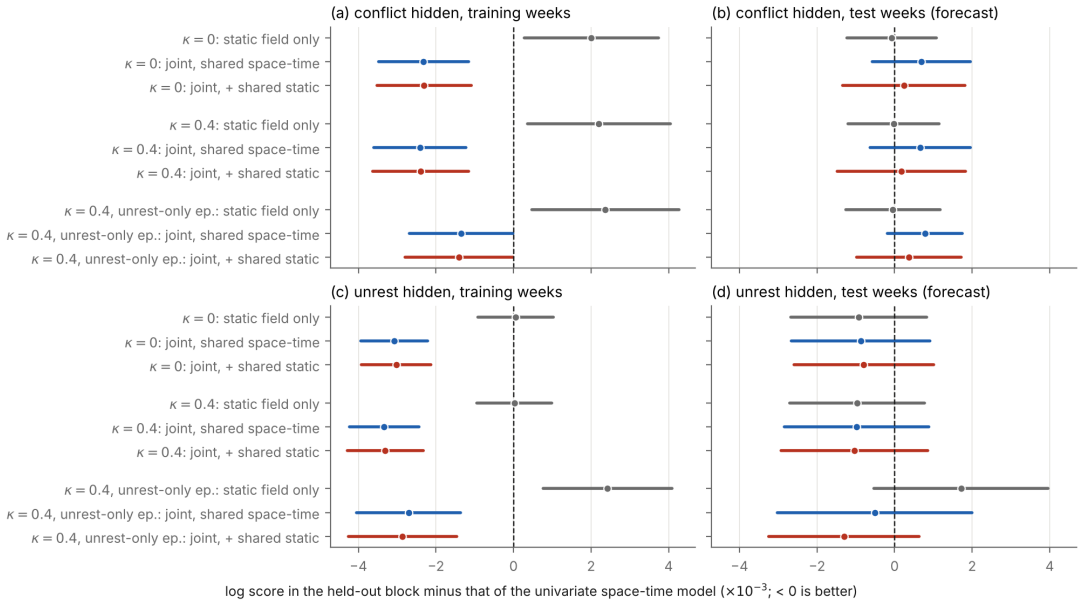


FIG 3. *S6: log score of the target in the held-out block, minus that of the univariate space-time model, paired by replicate, with conflict (a, b) or unrest (c, d) removed in the block; training weeks (a, c) and test weeks (b, d). Dots are means over 20 replicates and lines are 95% intervals (mean* ± 1.96 *SE). Grey: univariate model with a static field only; blue: joint model with a shared space-time field; red: joint model with shared space-time and static fields. Negative values favour the model. The extended joint model is within* 0.0005 *of the blue model throughout and is not shown.*

by at most 0.0003 in log score (SE 0.0005). The joint model should help when one outcome is missing in some area and the other is not, for example where conflict events go unreported but protests are recorded, or the reverse. S6 tests this.

Design.. The data come from the S5 generator, in the scenarios without escalation, with escalation, and with escalation and unrest-only episodes, with 20 replicates each. In each replicate a random block of 6×6 cells (16% of the domain) is chosen, and one outcome, the target, is removed there in all 130 weeks; the other outcome stays observed everywhere. Each target, conflict and unrest, is studied separately. Since the target's own history is then unknown in the block, its history covariate is set there to its mean over the observed rows, for every model. The univariate models of the target keep the history of the other outcome as a covariate, so they see the same data as the joint models, and any gain of the joint models comes from the latent fields they share. All models, hyperparameters included, are refitted on the remaining data:

- univariate models of the target with a static field, and with a static field and a space-time AR(1) field;
- the joint model with a shared space-time field of S5, and the extended model;
- the joint model with, in addition, a shared static field that enters the second outcome with its own loading, the structure of the generator.

The target is predicted in the block in the 104 training weeks, where it is unreported but the other outcome is observed in the same weeks, and in the 26 test weeks, a forecast in which the other outcome enters only through its recent history. The block holds on average 131 conflict and 160 unrest events in the training weeks and 33 and 42 in the test weeks.

Results.. Figure 3 shows the paired differences.

- *Training weeks.* The joint models predict the hidden outcome better than both univariate models in every scenario. Against the univariate space-time model the log score improves by 0.0020 for conflict and 0.0030 for unrest on average (SE 0.0003–0.0004), about 1.5% and 2% of the log score, and it improves in 80–85% and 92–95% of the replicates. AUC-PR rises by 0.006–0.008 and 0.011–0.014 (SE 0.002). Against the static-field model the gains are larger, 0.003–0.005 in log score and 0.017–0.029 in AUC-PR.
- *Test weeks.* The gain mostly disappears. For conflict the joint models are slightly worse than the univariate space-time model ($+0.0003$ to $+0.0007$, SE 0.0003–0.0004). For unrest they are better by 0.0008–0.0010 (SE 0.0005–0.0006), and in the scenario with unrest-only episodes the model with a shared static field improves on the static-field model by 0.0030 (SE 0.0009).

The joint models therefore borrow mainly what the other outcome shows about the current episode, in the same weeks. That information fills in an unreported area, but it decays in a forecast at the rate of the shared AR(1) field. The shared static field keeps only a small part of it; a plausible explanation is that, with events this rare, the persistent component is weak relative to the episodes (SD 0.6 against 0.8 here) and is already partly captured by the other outcome's history covariate. A fit takes a median of 135 seconds for the joint model with one shared field and 307 seconds with the shared static field, against 40 seconds for the univariate space-time model, on four cores.

5. Application to ACLED 2018–2024. Models are trained on 2018–2022 and evaluated on 2023–2024. The table is the repaired version of the cell-week table of Menicali, Marcantonio and Castruccio (2026); the repair, and the boosting analysis before and after it, are described in Sections S4.1 and S4.3 of the Supplementary Material. The computations were run at the Notre Dame Center for Research Computing.

5.1. *Escalation versus shared episodes.* Recent unrest predicts conflict: in every univariate model of conflict fitted here, the coefficient of the unrest history is positive and more than two standard errors from zero. S5 (Section 4.1) showed that this is weak evidence of escalation, because univariate models report escalation that does not exist when the two outcomes share latent episodes, and that the joint model recovers κ. The application differs from S5 in using steps of 13 weeks on a 1° lattice, cross-lags in both directions and the neighbour histories of both outcomes.

Region and models.. The region is West Africa and the Sahel, longitude 18°W to 16°E and latitude 4°N to 25°N. It has 2,505 cells in 22 countries and 906,810 cell-weeks per outcome with complete history. It contains the central Sahel, the Lake Chad basin and the coastal states of West Africa. The shared field v is defined on 641 cells of The shared field v is defined on 641 cells of 1°, with 28 steps of $\Delta = 13$ weeks. We fit four models, with the covariates, initial-conditions term, year effects and missingness indicators of Section S4.6 of the Supplementary Material, and the own and neighbour histories of both outcomes:

- *univariate static*: conflict only, with country effects, cell effects and a static Matérn field;
- *univariate space-time*: the same, plus a space-time field for conflict alone, which is the joint model with $\lambda = 0$;
- *joint*: model (3.4), in which the space-time field is shared by the two outcomes;
- *extended joint*: the joint model plus a space-time field for unrest alone.

Table 1 reports the lag coefficients, the hyperparameters of the space-time fields and the test scores; Table S18 of the Supplementary Material gives all hyperparameters and coefficients.

TABLE 1

Conflict and unrest in West Africa and the Sahel: lag coefficients (standard errors) of the conflict equation, hyperparameters of the space-time field, and test scores for 2023–2024. Own: conflict in the previous four weeks; other: unrest in the previous four weeks (κ); neighbour terms: the same histories averaged over the rook neighbours of the cell. In the joint model the space-time field is shared by the two outcomes and enters unrest with loading λ; its AR(1) coefficient is per step of 13 weeks. Full results in Table S18 of the Supplementary Material. Test scores here use the probit approximation to $E\{\mathrm{expit}(\eta)\}$; the model comparisons in the text use common Monte Carlo draws.

Region and model	$\hat{\kappa}$ unrest history	$\hat{\kappa}^{\mathrm{nb}}$ neighbours' unrest	$\hat{\delta}_{\mathrm{C}}^{\mathrm{nb}}$ neighbours' conflict	$\hat{\lambda}$	$\hat{\rho}_v$ (km)	$\hat{\phi}$
West Africa and Sahel						
Conflict only, $u + e + w$	0.138 (0.031)	0.005 (0.008)	0.129 (0.007)	–	–	–
Conflict only, $u + e + w + v$	0.131 (0.032)	0.003 (0.008)	0.051 (0.008)	–	423	0.90
Joint, shared v	0.044 (0.032)	−0.020 (0.008)	0.061 (0.008)	0.87	499	0.95
Joint, shared v + unrest-specific v_{U}	0.095 (0.032)	−0.005 (0.008)	0.054 (0.008)	0.58	445	0.96
AME region						
Conflict only, $u + e + w$	0.150 (0.014)	−0.004 (0.006)	0.175 (0.004)	–	–	–
Conflict only, $u + e + w + v$	0.133 (0.015)	−0.007 (0.006)	0.070 (0.004)	–	480	0.87
Joint, shared v	0.057 (0.015)	−0.035 (0.006)	0.083 (0.004)	0.67	486	0.95
Joint, shared v + unrest-specific v_{U}	0.105 (0.015)	−0.015 (0.006)	0.073 (0.004)	0.44	472	0.95

Results.. In the univariate static model the coefficient of recent unrest on conflict is $\hat{\kappa} = 0.138$ (SE 0.031). Adding a space-time field for conflict hardly changes it (0.131, SE 0.032), although the field is clearly present (range 423 km, standard deviation 0.69, AR(1) coefficient 0.90 per 13 weeks) and it reduces the own-lag coefficient from 0.575 (SE 0.016) to 0.444 (SE 0.017). This is the pattern of S5: a field estimated from conflict alone absorbs part of the persistence of conflict, but not the information about the current episode that is carried by recent unrest. In the joint model the coefficient falls to 0.044 (SE 0.032). The plug-in interval of ± 1.96 SE runs from -0.018 to 0.106, but it is likely too short. In S5 without escalation, the empirical SD of $\hat{\kappa}$ from the joint model was 1.20 times its mean SE, and in S3 the corresponding ratio for $\hat{\delta}$ was 1.19 to 1.66 whenever a space-time field was fitted (Sections 4.1 and S3.5). The own-lag coefficient stays at the level of the univariate space-time model (0.459, SE 0.017). The coefficient of unrest in neighbouring cells is close to zero in the univariate models (0.005 and 0.003, SE 0.008) and negative in the joint model (-0.020, SE 0.008).

The shared field has loading $\hat{\lambda} = 0.87$ on unrest, range 499 km, standard deviation 1.01 and AR(1) coefficient 0.95 per 13 weeks. It is a shared latent component of recorded risk that varies over several hundred kilometres and changes slowly over quarters; what it represents is discussed below. In the unrest equation of the joint model, the coefficient of recent conflict on unrest is 0.066 (SE 0.031), and that of conflict in the neighbouring cells is -0.042 (SE 0.013). Negative neighbour coefficients in both equations would mean that events in neighbouring cells lower the risk in a cell, which is implausible as a substantive spillover. One possible reading is that the shared field over-absorbs regional variation, and the neighbour terms compensate. We take them as a possible sign of misspecification of the joint model.

Interpretation.. Under the joint model with one shared field, the univariate association between recent unrest and conflict in this region is largely accounted for by shared episodes. The fall from 0.131 to 0.044 has two readings, however. The shared field has a larger standard deviation and range than the conflict-only field (1.01 against 0.69, and 499 against 423 km), which is what removing confounding by shared episodes would produce, but also what a field that absorbs unrest-only episodes would produce, with $\hat{\kappa}$ attenuated towards zero. The extended model separates them. It raises the log marginal likelihood by 131, its unrest-specific

field is substantial (range 787 km, SD 0.49, AR(1) coefficient 0.69 per 13 weeks), the loading of the shared field falls from 0.87 to 0.58, and $\hat{\kappa}$ rises to 0.095 (SE 0.032). In S5 the extended model gave slightly larger estimates than the model with one shared field, so part of this rise may reflect the model rather than the data.

Over Africa and the Middle East (13,719 cells in 93 countries, 4.9 million cell-weeks per outcome, v on 3,637 cells of 1° at 26-week steps) the pattern repeats with more precision: $\hat{\kappa} = 0.150$ (SE 0.014) in the univariate static model, 0.133 (0.015) in the univariate space-time model, 0.057 (0.015) in the joint model ($\hat{\lambda} = 0.67$) and 0.105 (0.015) in the extended model, which raises the log marginal likelihood by 692 and has a shared loading of 0.44. Once both shared and unrest-specific episodes are modelled, the short-run effect of recent unrest on conflict is about 0.10 in both regions, against 0.13–0.15 in the univariate models: shared episodes account for about a quarter to a third of the univariate association. [TODO: Model with a shared static field (biv3): West Africa $\hat{\kappa} = 0.061$ (SE 0.032), log marginal likelihood $+156$ over the extended model; add AME when job 1711845.2 finishes, and decide which model to report.]

Three limits apply. First, κ is the short-run, local part of escalation: escalation that unfolds over several months and hundreds of kilometres cannot be told apart from a shared episode and is attributed to v (Section 3.4). The finding is thus compatible with accounts in which repression of protest leads to armed conflict over longer periods (Ritter and Conrad, 2016; Ryckman, 2020; Chenoweth, Perkoski and Kang, 2017; Rød and Weidmann, 2023); it indicates that four-week unrest histories in the same cell are a weak signal of such escalation once shared episodes are modelled. Second, this is also why unrest histories help forecasting models of conflict (Rød, Hegre and Leis, 2025): they are informative about the current episode, whatever its nature. Third, the shared episodes mix occurrence and reporting, so neither v nor λ should be read as a property of occurrence alone.

5.2. *Prediction where one outcome is unobserved.* When both outcomes are observed, the joint model forecasts neither one better than the univariate models (Section S4.2 of the Supplementary Material), as expected. As in S6 (Section 4.2), we therefore ask whether it predicts an outcome better where that outcome is not observed but the other one is.

Design.. Countries with at least 20 events of the target outcome are split into five folds at random. In each fold the target is removed in the held-out countries in every week, 2018–2024, while the other outcome stays observed everywhere; its own history covariates are set there to their training mean, zero. The univariate models of the target keep the history of the other outcome, so they see the same data as the joint models. All models are refitted in every fold, hyperparameters included, with the case-control subsampling of Section 3.5. The target is scored in the held-out countries in the training years, where it is unreported but the other outcome is observed, and in the test years, a forecast. Standard errors are clustered by country.

Results.. In West Africa, with conflict as the target, the joint model predicts conflict in the held-out countries better than the univariate space-time model in both periods: the log score improves by 0.0036 (SE 0.0016) in the training years and by 0.0042 (SE 0.0021) in the test years, about 3% of the log score in both. In the test years, however, the univariate space-time model forecasts worse than the univariate static model (log score 0.152 against 0.147), and against the static model the joint model improves only in the training years, by 0.0031 (SE 0.0022), not in the test years ($+0.0008$, SE 0.0023). The gains are largest in Nigeria, Cameroon and Morocco in the training years, and Nigeria turns into a loss in the test years. The extended joint model gives similar results. With unrest as the target there is no gain: -0.0003 (SE 0.0012) in the training years and $+0.0031$ (SE 0.0015) in the test years against

the univariate static model, the loss coming almost entirely from Ghana. [TODO: Africa and the Middle East (job 1712445, tasks 11–20); with the hyperparameters of the full fit, the joint model improved the log score of conflict in held-out countries by 0.0023 (SE 0.0012) in the training years and 0.0018 (SE 0.0015) in the test years, with the largest gains in Palestine, Lebanon, Yemen, Syria and Afghanistan. Add a figure of gains by country.]

As in S6, the joint model helps most in filling in an outcome that is unreported in some area while the other is recorded, and less in forecasting it. Where conflict reporting is sparse but protests are recorded, or the reverse, the shared field transfers information on the current episode from one outcome to the other.

6. Discussion.

[TODO: Discussion (not drafted yet).]

Acknowledgments. *Use of generative AI.* This work is AI-assisted. The research questions, the choice of models and of simulation designs, the interpretation of the results and all decisions about the analysis were made by the authors. Claude, a large language model developed by Anthropic, accessed through the Claude Code tool between October and November 2026, was used as an assistant under the authors' direction: (i) to write and test computer code, including the latent Gaussian model and its Laplace-approximation inference engine, the simulator and the simulation studies, the reconstruction of the cell-week data set, the reproduction of the earlier boosting analysis, and the scripts that produce the tables and figures; (ii) to help check the earlier data-processing pipeline, which led to the errors reported in Section 5; (iii) to help draft and edit the text and compile the bibliography; and (iv) to carry out internal reviews of the draft before submission. The authors checked the code, results, text and references and take full responsibility for the content of the paper. Claude is not an author of this work. All code is available in the supplementary material.

Funding. [TODO: Funding.]

SUPPLEMENTARY MATERIAL

Code

Python implementation of the latent Gaussian model, the simulator and all simulation studies.

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